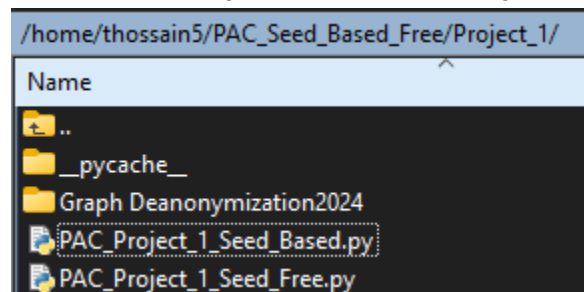


## Steps (Seed Based, Seed Free Graph Deanonymization):

1. Create a Directory **Project\_1**
  - a. Keep **Graph Deanonymization2024** Directory in it, the directory contains the following folders
    - i. Seed\_based
    - ii. Seed\_free
    - iii. validation\_Dataset
  - b. Keep two Python files
    - i. **PAC\_Project\_1\_Seed\_Based.py**
    - ii. **PAC\_Project\_1\_Seed\_Free.py**



2. Next in the command line interface (CLI), Navigate to the **Project\_1** directory
  - a. **cd Project\_1**
3. Then execute the .py files with commands
  - a. **python PAC\_Project\_1\_Seed\_Based.py**
  - b. **python PAC\_Project\_1\_Seed\_Free.py**
4. If you want to **change arguments** then you have to use the following commands
  - a. For Seed Based
    - i. **python PAC\_Project\_1\_Seed\_Based.py --path <Project\_1 Directory's address> --train\_portion <Percentage of Given Seed Pairs form validation> --ecce\_threshold <Eccentricity Threshold>**
    - ii. The output file you get: **Tanvir\_project\_1\_final\_mapping\_seed\_based.txt**

```
(Seed GraphDeAnon) thossain5@deliab:~/PAC_Seed_Based_Free/Project_1$ python PAC_Project_1_Seed_Based.py --path /home/thossain5/PAC_Seed_Based_Free/Project_1 --train_portion 0.5 --ecce_threshold 0.2
Current Working Directory: /home/thossain5/PAC_Seed_Based_Free/Project_1
/home/thossain5/PAC_Seed_Based_Free/Project_1/Graph Deanonymization2024/validation_Dataset
Graph 1: Graph with 26956 nodes and 344375 edges
Graph 2: Graph with 27756 nodes and 348903 edges

Check lengths:
seed pair 5388

seed nodes in graph 1: 5388
seed nodes in graph 2: 5388

A new Pair has been added: (27358, 20105):0.7142857142857142
A new Pair has been added: (25194, 26998):0.7071067811865475
A new Pair has been added: (7474, 14900):0.17677669529663687
A new Pair has been added: (27330, 15448):0.9999999999999999
A new Pair has been added: (18606, 25215):0.5000000000000001
A Pair is going to be deleted for 14900: (7474,14900):0.17677669529663687
A new Pair has been added: (7468, 14900):0.9999999999999999
```

- b. For Seed Free
        - i. **python PAC\_Project\_1\_Seed\_Free.py --method\_name <name of the centrality measure method such as 'pagerank' or 'eigen' or 'degree'> --path <Project\_1 Directory's address> --integrate\_with\_Seed\_Based <**

0 or 1 (If you want to add top k pairs of your method with Seed Based method)> **--number\_of\_top\_index\_pairs** <Before integrating Seed Based method how many (> 0) top indices pairs you want to provide as given seed pairs, automatically assign zero (0) when we do not integrate Seed Based method>

- ii. Note that if we do not --integrate\_with\_Seed\_Based(=0) then --number\_of\_top\_index\_pairs is set as zero
- iii. If Dataset graphs are very sparse please do not call 'pagerank' it will show you the iteration depth error
- iv. The output file you get:
  1. **Tanvir\_Project\_1\_final\_mapping\_seed\_free\_1.txt** ( when you integrate with the Seed-based method )
  2. **Tanvir\_Project\_1\_final\_mapping\_seed\_free\_0.txt** ( when you do not integrate with the Seed-based method

```

[Seed_GraphDeAnon] thossain5@celab:~/PAC_Seed_Based_Free/Project_1$ python PAC_Project_1_Seed_Free.py --method_name 'pagerank' --path /home/thossain5/PAC_Seed_Based_Free/Project_1 --integrate_with_Seed_Based 1 --number_of_top_index_pairs 50
PyTorch version: 2.5.1+cu124
CUDA version: 12.4
Current Working Directory: /home/thossain5/PAC_Seed_Based_Free/Project_1
/home/thossain5/PAC_Seed_Based_Free/Project_1/Graph_DeAnonymization2024/validation_Dataset
(1556, 108800) 0 0, ('124', '2846') 1 1, ('718', '177149') 2 2, ('37', '4089') 3 3, ('111', '22231') 4 4, ('589', '801') 5 5, ('468', '15496') 6 6, ('250', '24477') 7 7, ('10', '14415') 8 8, ('8', '12146') 9 9, ('411', '838') 10 10, ('508', '1589')
11 11, ('246', '22510') 12 12, ('268', '22968') 13 13, ('443', '23187') 14 14, ('155', '15789') 15 15, ('302', '13070') 16 16, ('50', '24929') 17 17, ('765', '6967') 18 18, ('15', '5232') 19 19, ('157', '9265') 20 20, ('305', '24031') 21 21, ('1192', '21259') 22 22, ('110', '24632') 23 23, ('688', '3858') 24 24, ('384', '22361') 25 25, ('160', '4303') 26 26, ('35', '24465') 27 27, ('335', '24800') 28 28, ('632', '25760') 29 29, ('2158', '13861') 30 30, ('1209', '18608') 31 31, ('166', '25050') 32 32, ('270', '24435') 33 33, ('154', '11507') 34 34, ('1522', '928') 35 35,
Page Rank:
Number of Extracted Seed Pairs: 36      Accuracy: 0.1336203496830228
seed nodes in graph 1: 50
seed nodes in graph 2: 50

A new Pair has been added: (10950, 4094) 10.1569596452595352
A new Pair has been added: (345, 21947) 10.10314212462557934
A new Pair has been added: (14236, 18448) 10.277380091122446
A new Pair has been added: (12152, 24277) 10.154303494620819
A new Pair has been added: (14719, 11928) 10.08512463507587489
A new Pair has been added: (24182, 13440) 10.3014113491777432

```

c. Output files like:

- i. **result\_0.2\_0.5.txt** - in this file, accuracy is reported for 0.2 or 20% training validation pairs and 0.5 eccentricity threshold.
- ii. **result\_0.txt** - in this file, accuracy is reported for the method having **no integration** to Seed Based method
- iii. **result\_1.txt** - in this file, accuracy is reported for the method **having integration** to Seed Based method
- iv. **The other three files (yellow marked)** - list all the corresponding node pairs obtained from our method

| /home/thossain5/PAC_Seed_Based_Free/Project_1/ |        |
|------------------------------------------------|--------|
| Name                                           | Size   |
| ..                                             |        |
| __pycache__                                    |        |
| Graph DeAnonymization2024                      |        |
| PAC_Project_1_Seed_Based.py                    | 15 KB  |
| PAC_Project_1_Seed_Free.py                     | 17 KB  |
| result_0.2_0.5.txt                             | 1 KB   |
| result_0.txt                                   | 1 KB   |
| result_50.txt                                  | 1 KB   |
| Tanvir_project_1_final_mapping_seed_based.txt  | 279 KB |
| Tanvir_Project_1_final_mapping_seed_free_0.txt | 295 KB |
| Tanvir_Project_1_final_mapping_seed_free_1.txt | 277 KB |